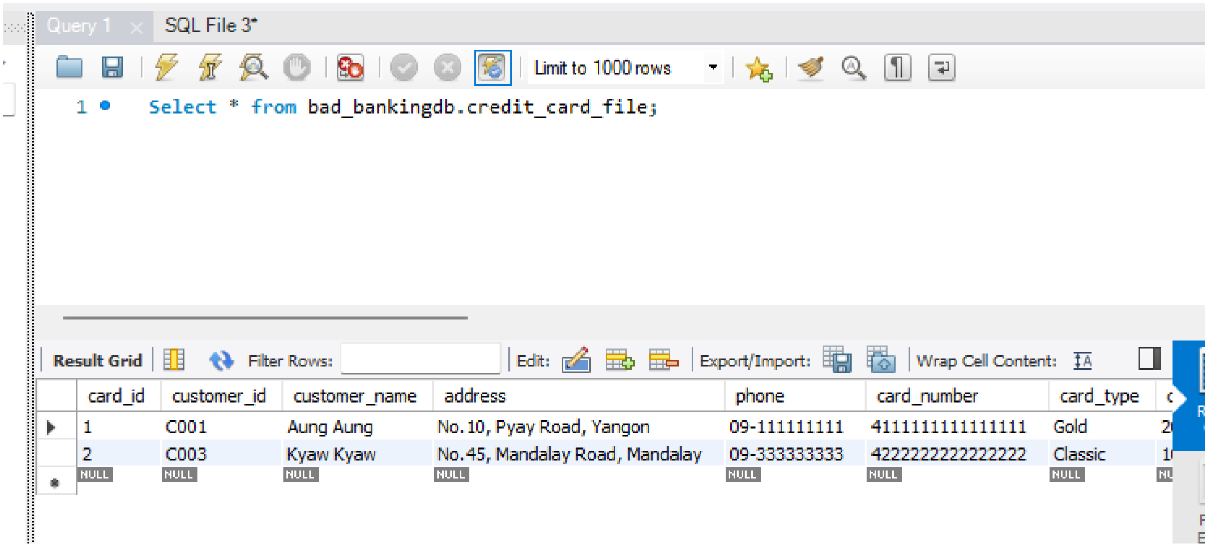
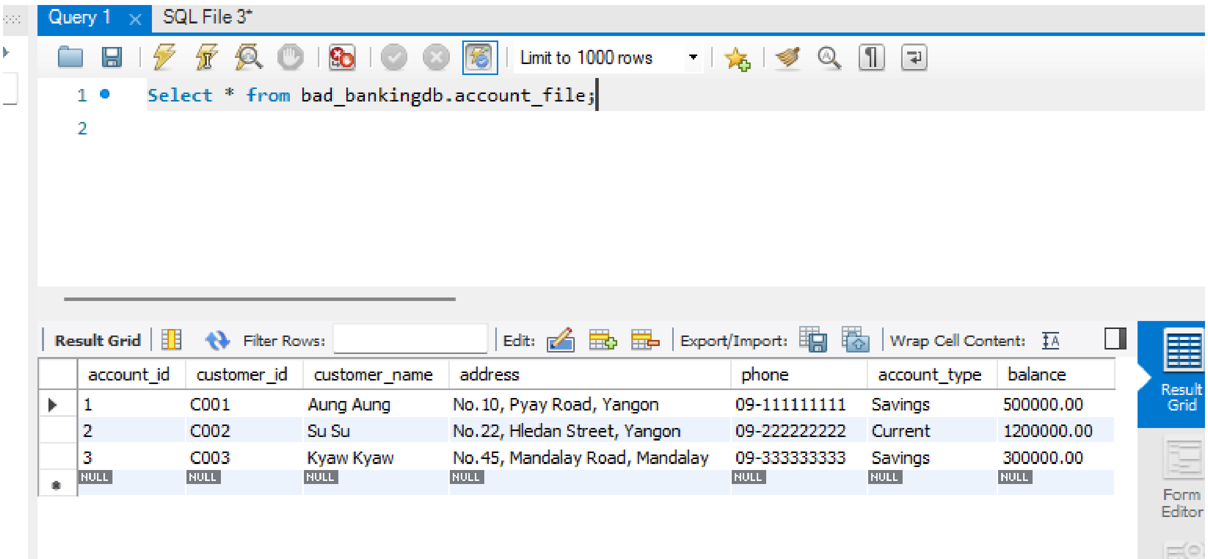
Bad Banking Database



```
Select * from bad_bankingdb.credit_card_file;
```

card_id	customer_id	customer_name	address	phone	card_number	card_type	c
1	C001	Aung Aung	No. 10, Pyay Road, Yangon	09-111111111	4111111111111111	Gold	2
2	C003	Kyaw Kyaw	No. 45, Mandalay Road, Mandalay	09-333333333	4222222222222222	Classic	1
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NU

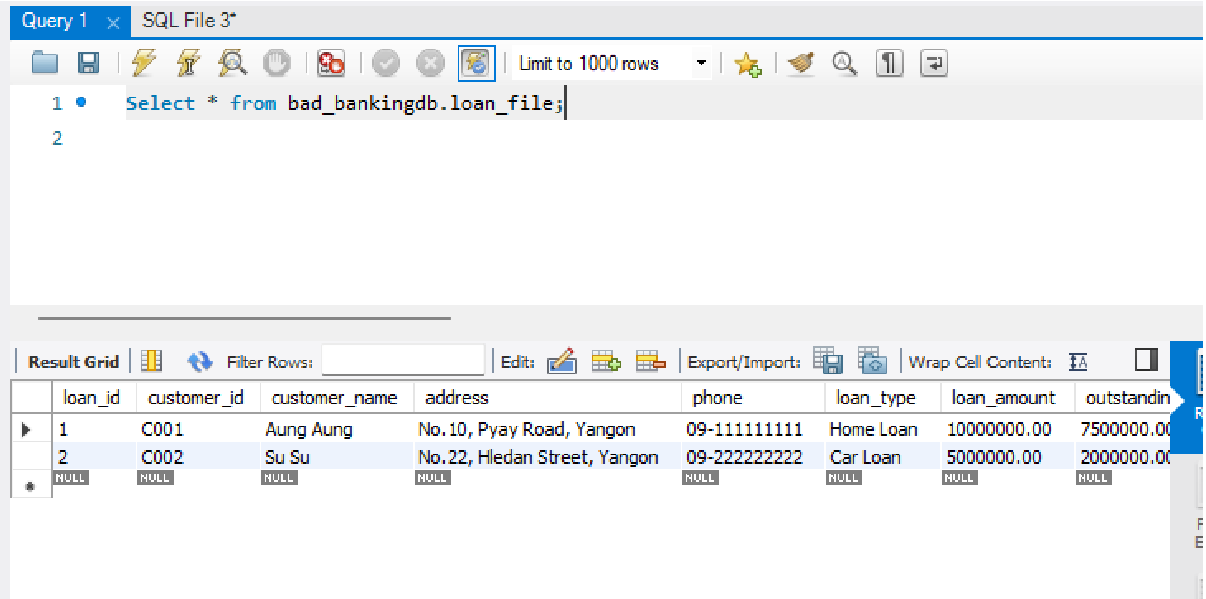
credit_card_files table



```
Select * from bad_bankingdb.account_file;
```

account_id	customer_id	customer_name	address	phone	account_type	balance
1	C001	Aung Aung	No. 10, Pyay Road, Yangon	09-111111111	Savings	500000.00
2	C002	Su Su	No. 22, Hledan Street, Yangon	09-222222222	Current	1200000.00
3	C003	Kyaw Kyaw	No. 45, Mandalay Road, Mandalay	09-333333333	Savings	300000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL

account_files table



loan_id	customer_id	customer_name	address	phone	loan_type	loan_amount	outstandin
1	C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	Home Loan	10000000.00	7500000.00
2	C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	Car Loan	5000000.00	2000000.00
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

loan_files table

==

-- WORKFLOW PROBLEM

-- Customer C001 changes address

-- Only account_file is updated

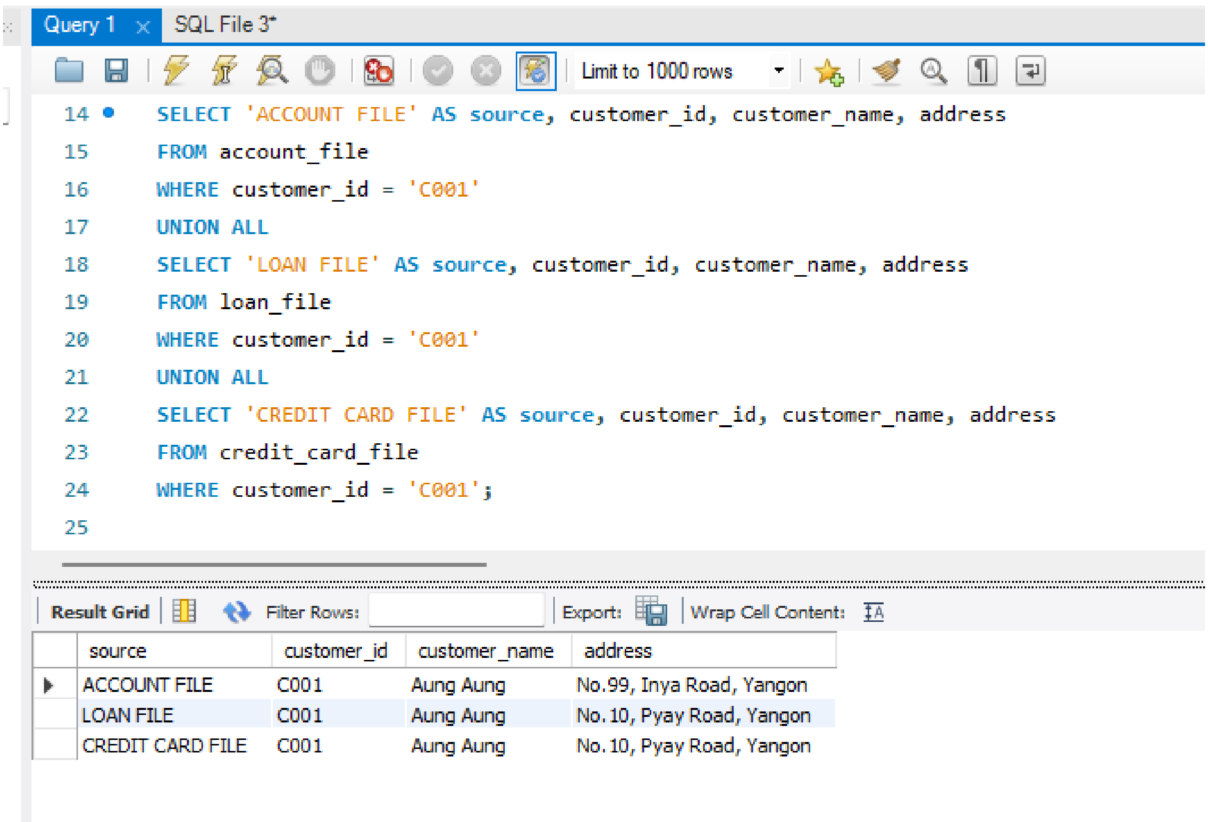
==

UPDATE account_file

SET address = 'No.99, Inya Road, Yangon'

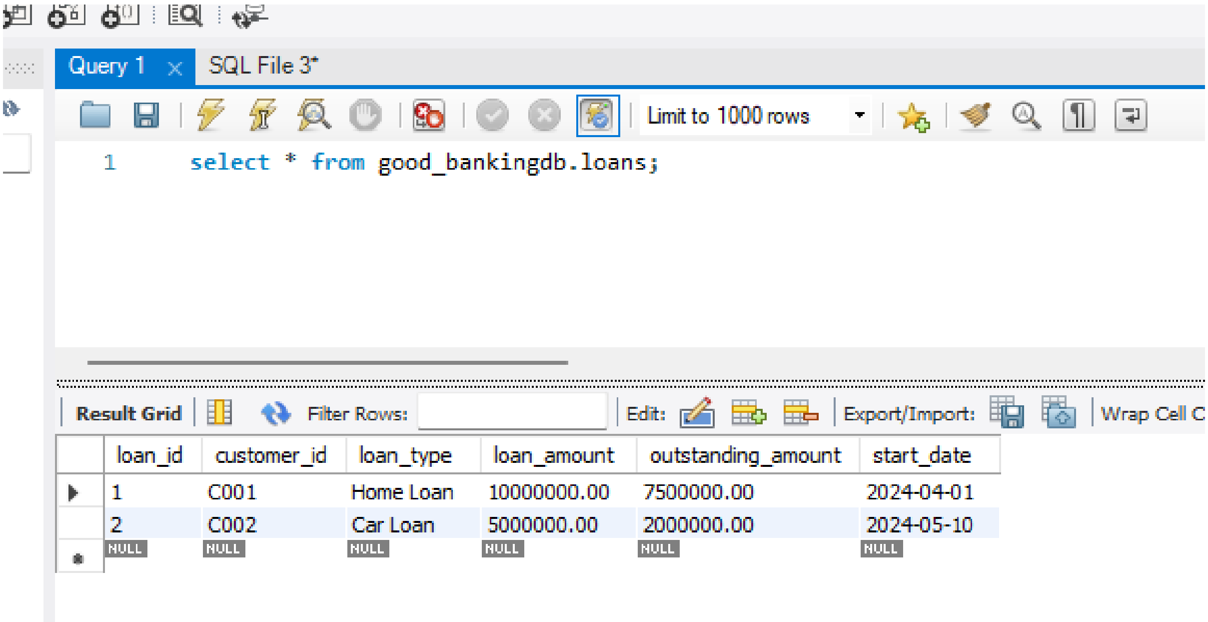
WHERE customer_id = 'C001';

After this update, let’s check consistency



Inconsistency data

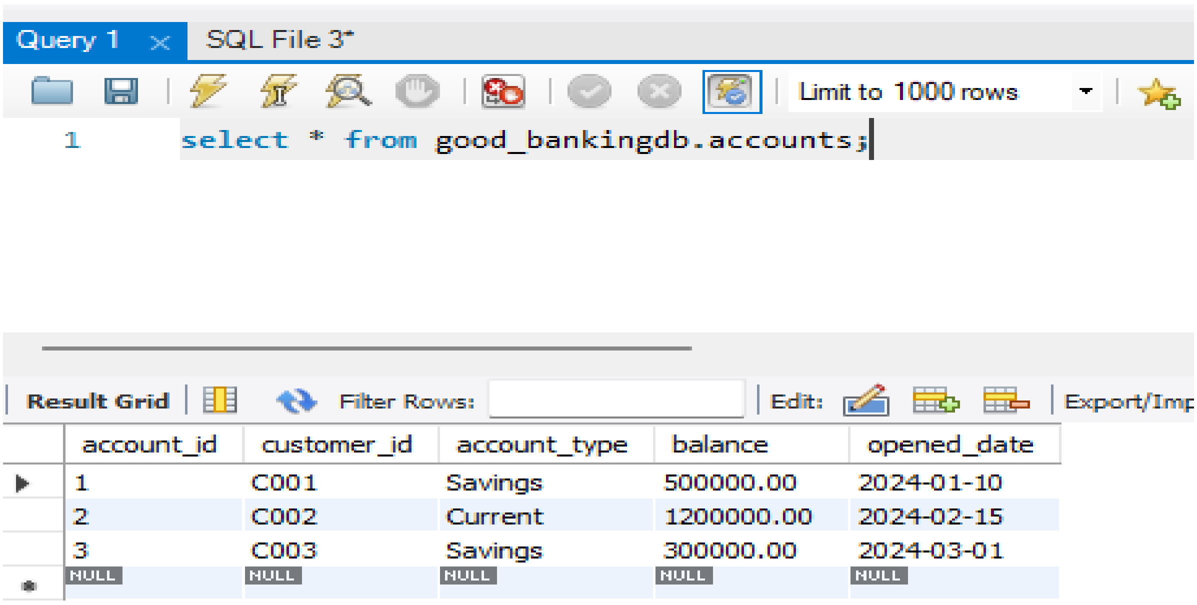
Good Banking Database



```
select * from good_bankingdb.loans;
```

loan_id	customer_id	loan_type	loan_amount	outstanding_amount	start_date
1	C001	Home Loan	10000000.00	7500000.00	2024-04-01
2	C002	Car Loan	5000000.00	2000000.00	2024-05-10
NULL	NULL	NULL	NULL	NULL	NULL

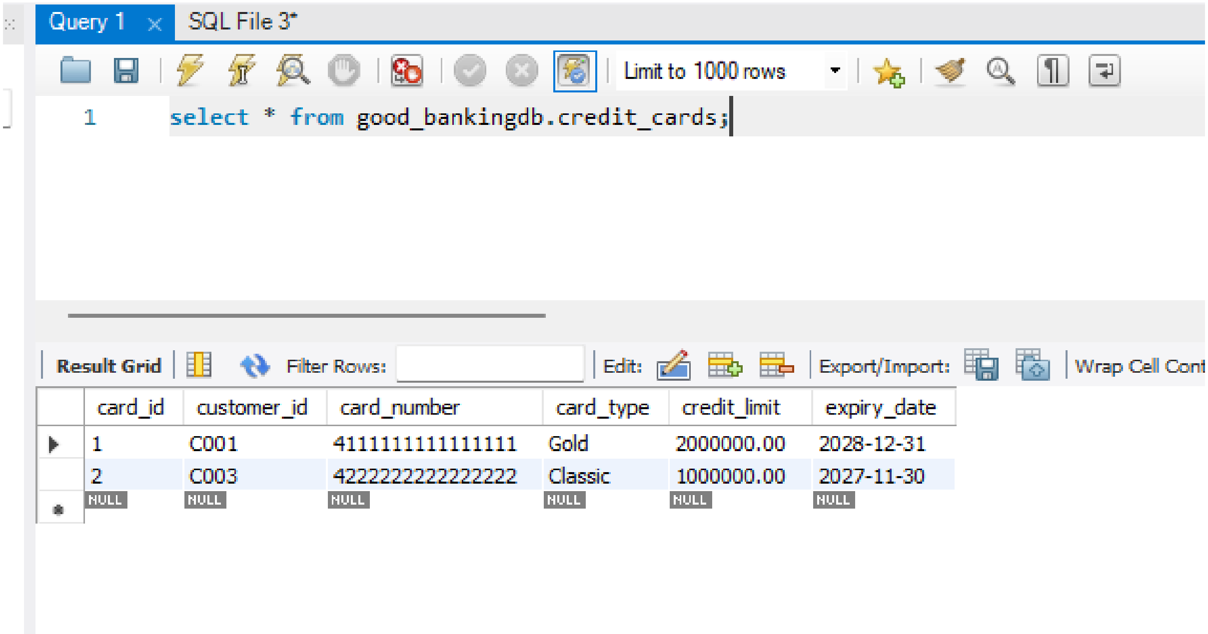
loans table



```
select * from good_bankingdb.accounts;
```

account_id	customer_id	account_type	balance	opened_date
1	C001	Savings	500000.00	2024-01-10
2	C002	Current	1200000.00	2024-02-15
3	C003	Savings	300000.00	2024-03-01
NULL	NULL	NULL	NULL	NULL

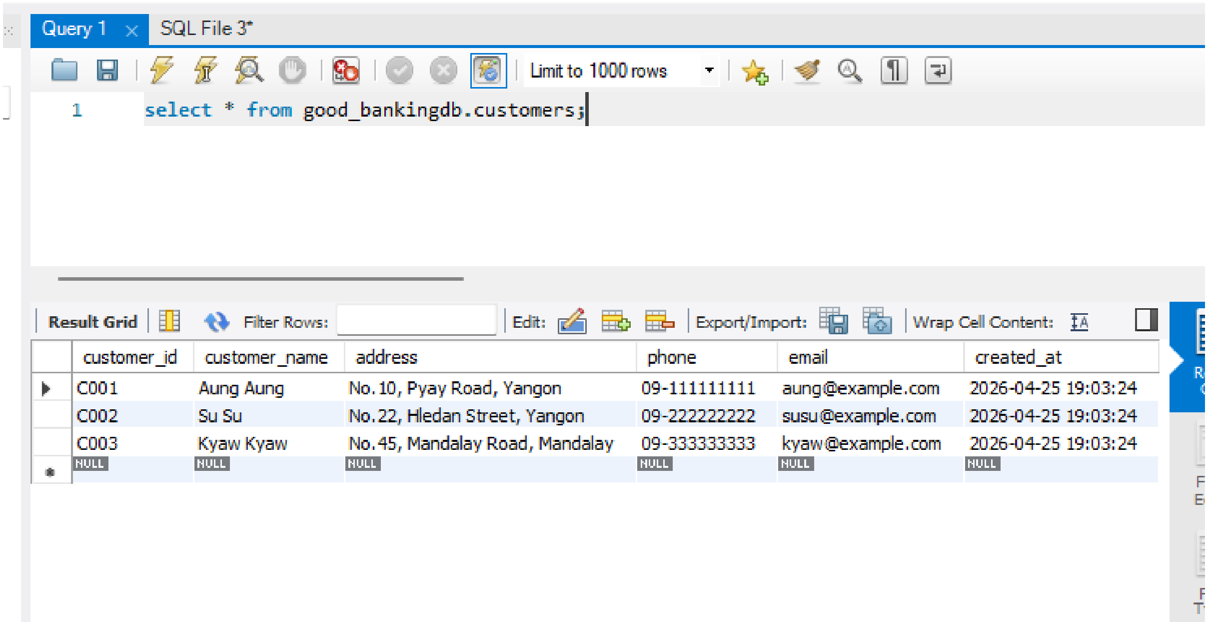
accounts table



```sql
select * from good_bankingdb.credit_cards;
```

card_id	customer_id	card_number	card_type	credit_limit	expiry_date
1	C001	4111111111111111	Gold	2000000.00	2028-12-31
2	C003	4222222222222222	Classic	1000000.00	2027-11-30
NULL	NULL	NULL	NULL	NULL	NULL

credit_cards table



```sql
select * from good_bankingdb.customers;
```

customer_id	customer_name	address	phone	email	created_at
C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	aung@example.com	2026-04-25 19:03:24
C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	susu@example.com	2026-04-25 19:03:24
C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	kyaw@example.com	2026-04-25 19:03:24
NULL	NULL	NULL	NULL	NULL	NULL

customomres table

UPDATE customers

SET address = 'No.99, Inya Road, Yangon'

WHERE customer_id = 'C001';

-- Check consistent result through joins

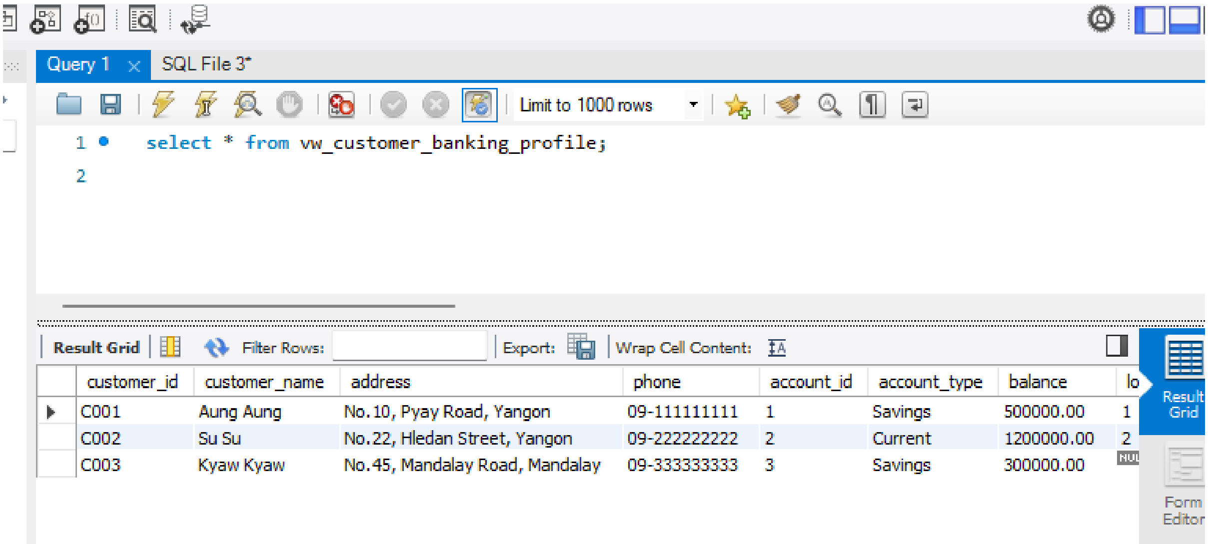
```sql
select * from good_bankingdb.customers;
SELECT
    c.customer_id,
    c.customer_name,
    c.address,
    a.account_type,
    a.balance,
    l.loan_type,
    l.outstanding_amount,
    cc.card_number,
    cc.card_type
FROM customers c
LEFT JOIN accounts a ON c.customer_id = a.customer_id
LEFT JOIN loans l ON c.customer_id = l.customer_id
LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
WHERE c.customer_id = 'C001';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_name	address	account_type	balance	loan_type	outstanding_amount	car
C001	Aung Aung	No. 10, Pyay Road, Yangon	Savings	500000.00	Home Loan	7500000.00	411

consistency data

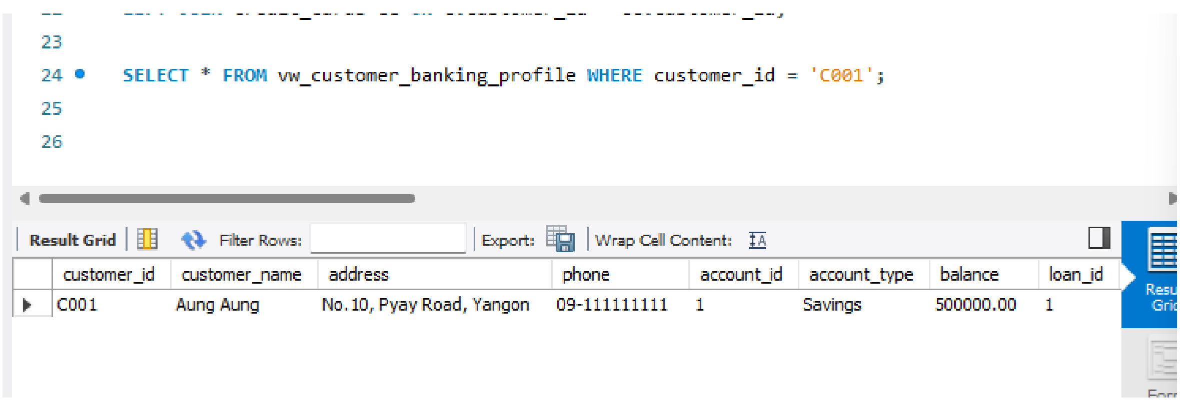
-- View for Easy Demonstration



```
select * from vw_customer_banking_profile;
```

customer_id	customer_name	address	phone	account_id	account_type	balance	lo
C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	1	Savings	500000.00	1
C002	Su Su	No.22, Hledan Street, Yangon	09-222222222	2	Current	1200000.00	2
C003	Kyaw Kyaw	No.45, Mandalay Road, Mandalay	09-333333333	3	Savings	300000.00	NULL

View table



```
SELECT * FROM vw_customer_banking_profile WHERE customer_id = 'C001';
```

customer_id	customer_name	address	phone	account_id	account_type	balance	loan_id
C001	Aung Aung	No.10, Pyay Road, Yangon	09-111111111	1	Savings	500000.00	1

Test Case

Test case (1) with bad_bankingdb

After updaing these (

UPDATE account_file

SET address = 'No.10, Pyay Road, Yangon'

WHERE customer_id = 'C001';

UPDATE loan_file

SET address = 'No.10, Pyay Road, Yangon'

WHERE customer_id = 'C001';

UPDATE credit_card_file

SET address = 'No.10, Pyay Road, Yangon'

WHERE customer_id = 'C001';

-- Update only account file

UPDATE account_file SET address = 'No.99, Inya Road, Yangon' WHERE customer_id = 'C001';

)

Check consistency

```sql
-- Verify inconsistency
SELECT 'ACCOUNT FILE' AS source, address
FROM account_file
WHERE customer_id = 'C001'
UNION ALL
SELECT 'LOAN FILE' AS source, address
FROM loan_file
WHERE customer_id = 'C001'
UNION ALL
SELECT 'CREDIT CARD FILE' AS source, address
FROM credit_card_file
WHERE customer_id = 'C001';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

source	address
ACCOUNT FILE	No.99, Inya Road, Yangon
LOAN FILE	No. 10, Pyay Road, Yangon
CREDIT CARD FILE	No. 10, Pyay Road, Yangon

Inconsistency Data

Test Case (2) with good_bankingdb

Run these (

-- Reset centralized data

UPDATE customers SET address = 'No.10, Pyay Road, Yangon' WHERE customer_id = 'C001';

-- Update once UPDATE customers SET address = 'No.99, Inya Road, Yangon' WHERE customer_id = 'C001';

)

-- Verify consistency

```
SELECT
    c.customer_id,
    c.customer_name,
    c.address,
    a.account_type,
    l.loan_type,
    cc.card_type
FROM customers c
LEFT JOIN accounts a ON c.customer_id = a.customer_id
LEFT JOIN loans l ON c.customer_id = l.customer_id
LEFT JOIN credit_cards cc ON c.customer_id = cc.customer_id
WHERE c.customer_id = 'C001';
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

customer_id	customer_name	address	account_type	loan_type	card_type
C001	Aung Aung	No.99, Inya Road, Yangon	Savings	Home Loan	Gold

Consistency Data